PQC DAILY STUDY SERIES

The Role of TRNGs and QRNGs in Post-Quantum Cryptography

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The Critical Role of Randomness in PQC: TRNGs and QRNGs

Post-Quantum Cryptography (PQC) introduces new mathematical foundations to resist attacks from quantum computers. However, the security of any cryptographic system, including PQC, fundamentally relies on the quality of its random numbers. This is where True Random Number Generators (TRNGs) and Quantum Random Number Generators (QRNGs) play a pivotal role.

Why is High-Quality Randomness So Important?

In cryptography, random numbers are essential for generating secret keys, nonces (numbers used once), and other values that must be unpredictable. If an attacker can guess or predict these numbers, the entire cryptographic system can be compromised, regardless of the strength of the underlying algorithm. PQC algorithms, such as ML-KEM (Kyber) and ML-DSA (Dilithium), require large, statistically uniform, and unpredictable random numbers for their key generation processes. A weak random number generator can introduce subtle biases that a sophisticated attacker could exploit.

> **Key takeaway:** The security of a PQC algorithm is not just about its mathematical hardness; it is equally about the unpredictability of the secrets it generates. Weak randomness undermines the entire system.

TRNGs vs. QRNGs: Sources of True Randomness

Unlike Pseudorandom Number Generators (PRNGs), which are deterministic algorithms that produce a sequence of numbers that appear random but are ultimately predictable if the initial "seed" is known, TRNGs and QRNGs derive randomness from physical phenomena.

- **True Random Number Generators (TRNGs):** These devices extract randomness from unpredictable physical processes like thermal noise, atmospheric noise, or the jitter in electronic circuits. They are designed to produce output that is statistically random and free from patterns.
- **Quantum Random Number Generators (QRNGs):** QRNGs leverage the inherent randomness of quantum mechanics. They measure quantum phenomena, such as the path a photon takes through a beam splitter or the timing of radioactive decay, which are fundamentally unpredictable according to the laws of physics. This provides the highest possible quality of randomness.

TRNG vs. QRNG Comparison

Feature	TRNG (True Random Number Generator)	QRNG (Quantum Random Number Generator)
Source of Randomness	Classical physical phenomena (e.g., thermal noise, clock jitter)	Quantum mechanical phenomena (e.g., photon polarization, quantum tunneling)
Theoretical Foundation	Based on the unpredictability of complex classical systems	Based on the fundamental indeterminacy of quantum physics
Security Assurance	High, but relies on the physical model being truly unpredictable	Highest, rooted in the foundational principles of quantum mechanics
Implementation Complexity	Varies, can be integrated into silicon chips	Generally more complex and expensive, but becoming more accessible

Applicability to Internet Protocols (TLS, SSH)

In protocols like TLS 1.3 and SSH, which are being updated to support PQC, the need for high-quality randomness is paramount during the handshake phase. When a client and server establish a secure connection, they generate ephemeral (short-lived) keys. In a hybrid PQC scheme, this involves generating keys for both a classical algorithm (like X25519) and a PQC algorithm (like ML-KEM). The randomness used to generate these keys must be robust. Using a TRNG or QRNG at the server and client endpoints ensures that these ephemeral keys are truly unpredictable, protecting the session against both classical and quantum adversaries.

Daily Quiz

1. What is the fundamental difference between a PRNG and a TRNG?
2. Why is a QRNG considered to provide a higher quality of randomness than a TRNG?
3. In the context of a TLS handshake using a hybrid PQC scheme, where is high-quality randomness most critical?